AQA 3.1.9 Rate equations

AQA A-level Chemistry · Revision questions · Sheet 3 of 6

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

Sheet 3 of 6

20 marks · about 25 minutes

Key information

- Units of k: rearrange k = rate ÷ (concentration terms), with rate in mol dm^{-3} s^{-1}, and cancel
- Rate-determining step = slowest step; species in the rate equation appear in (or before) it
- $\ln k = -E_a/RT + \ln A$, so the gradient of ln k against 1/T = $-E_a/R$
- Clock reactions: the same amount of product forms each time, so initial rate $\propto 1/t$
- rate = $k[A]^m[B]^n$ — orders come from experimental data, not from the balanced equation

Section A: Short-answer questions

01 Nitrogen dioxide reacts with carbon monoxide in a two-step mechanism.

Step 1 $NO_2 + NO_2 \rightarrow NO_3 + NO$ (slow)

Step 2 $NO_3 + CO \rightarrow NO_2 + CO_2$ (fast)

Deduce the rate equation for the overall reaction.

[1 mark]

02 The results of an investigation of the reaction between P and Q are shown in this table.

Experiment	Initial [P] / mol dm^{-3}	Initial [Q] / mol dm^{-3}	Initial rate / mol dm^{-3} s^{-1}
1	0.200	0.500	0.400
2	0.600	To be calculated	0.800

The rate equation is: rate = $k[P][Q]^2$

What is the initial concentration of Q in experiment 2?

[1 mark]

A 0.167
B 0.333
C 0.408
D 0.612

03 The rate equation for a reaction is rate = $k[A]^2[B]$. State the effect on the rate of halving the concentration of A only.

[1 mark]

04 When the concentration of reactant E is doubled, with all other conditions kept the same, the initial rate of reaction increases by a factor of 4. Deduce the order of reaction with respect to E.

[1 mark]

05 The rate equation for the acid-catalysed reaction between iodine and propanone is: rate = $k[H^+][C_3H_6O]$

The rate of reaction was measured for a mixture of iodine, propanone and sulfuric acid at pH = 0.70. In a second mixture the concentration of the sulfuric acid was different but the concentrations of iodine and propanone were unchanged. The new rate of reaction was a quarter of the original rate.

What was the pH of the second mixture?

[1 mark]

A 1.00
B 1.30
C 1.40
D 2.80

06 The rate equation for a reaction is rate = $k[A]^2[C]$. Give the overall order of this reaction.

[1 mark]

Section B: Longer questions

07 Bromate(V) ions and bromide ions react in acid conditions according to the equation

$BrO_3^-(aq) + 5Br^-(aq) + 6H^+(aq) \rightarrow 3Br_2(aq) + 3H_2O(l)$

A series of experiments was carried out at a given temperature. The results were used to deduce the rate equation for the reaction.

rate = $k[BrO_3^-][Br^-][H^+]^2$

Table 1 shows an incomplete set of results.

Experiment	Initial $[BrO_3^-]$ / mol dm^{-3}	Initial $[Br^-]$ / mol dm^{-3}	Initial $[H^+]$ / mol dm^{-3}	Initial rate / mol dm^{-3} s^{-1}
1	0.10	0.20	0.30	2.4×10^{-2}
2		0.20	0.30	3.6×10^{-2}
3	0.20	0.40	0.50	
4	0.10	0.10		2.7×10^{-2}

07.1 Use the data from Experiment 1 to calculate a value for the rate constant, k, at this temperature and give its units. Give your answer to an appropriate number of significant figures.

[3 marks]

07.2 Complete Table 1.

[3 marks]

08 The effect of reactant concentrations on the rate of a reaction is investigated.

$H_2O_2(aq) + 2I^-(aq) + 2H^+(aq) \rightarrow I_2(aq) + 2H_2O(l)$

The experiment is done five times in the presence of thiosulfate ions and starch. The reactant concentrations are varied by changing the volume of each reactant in the mixture. The thiosulfate ions react with the iodine as it is produced, until all the thiosulfate has reacted: $I_2(aq) + 2S_2O_3^{2-}(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$

After all the thiosulfate has reacted, the iodine produced then reacts with the starch and the solution turns blue-black. The time taken for the solution to turn blue-black is recorded. If the amount of thiosulfate is the same in each experiment, the time recorded represents the time to produce the same amount of iodine in each experiment.

Experiment	$[H_2O_2]$ / mol dm^{-3}	$[I^-]$ / mol dm^{-3}	$[H^+]$ / mol dm^{-3}	Time to turn blue-black / s
1	0.010	0.010	0.10	60
2	0.020	0.010	0.10	30
3	0.030	0.010	0.10	20
4	0.030	0.020	0.10	10
5	0.030	0.020	0.20	10

08.1 The amount of thiosulfate is the same in each experiment. Identify three other control variables.

[3 marks]

08.2 Deduce the order of reaction with respect to each reactant. Explain how you deduced the order of reaction with respect to iodide ions.

[3 marks]

08.3 Give the rate equation for the reaction. Determine the units of the rate constant.

[2 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	rate = $k[NO_2]^2$	*Source: Practice question (written for this set)* CO not in rate equation as it is only involved after the slow step	1
02	**C**	*Source: 7405/3 Jun 2019 Q10*	1
03	Rate falls to ¼ / quartered	*Source: Practice question (written for this set)*	1
04	Second order / 2	*Source: Practice question (written for this set)*	1
05	**B**	*Source: 7405/3 Jun 2017 Q27*	1
06	3 / third order	*Source: Practice question (written for this set)*	1
07.1	M1 $k = 2.4 \times 10^{-2} \div (0.10 \times 0.20 \times 0.30^2)$ (= 13.333) M2 = 13 (must be 2 sf) M3 units $mol^{-3}\ dm^9\ s^{-1}$	*Source: 7405/2 Jun 2018 Q05.1–05.2* M1 is for insertion of numbers into a correctly rearranged equation; units in any order	3
07.2	Experiment 2 $[BrO_3^-] = 0.15$ Experiment 3 rate = 0.26 or 0.27 Experiment 4 $[H^+] = 0.45$ or 0.46	If k wrong in first part, allow expected answers or values consequential on their k	3
08.1	Any three from: temperature; amount / volume of starch; total volume of reaction mixture; (constant) stirring / agitation; same person observing / same shade of colour when timing stopped	*Source: 7405/3 Jun 2025 Q02.1–02.3* Ignore same stopwatch Ignore concentrations / volumes of thiosulfate and other reagents	3
08.2	M1 first order for H_2O_2 **and** first order for I^- **and** zero order for H^+ M2 doubling $[I^-]$ halves the time M3 (using experiments 3 and 4) because $[H^+]$ and $[H_2O_2]$ stay the same / only $[I^-]$ changes	M2 allow doubling $[I^-]$ doubles the rate	3
08.3	M1 rate = $k[H_2O_2][I^-]$ M2 $mol^{-1}\ dm^3\ s^{-1}$	M2 dependent on a rate equation in M1	2